CV - Daniel Childs

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Consultant Software Engineer with 20 years' experience delivering software across startups, consultancies, and global enterprises. Trusted full-stack technical leader, specialising in cloud-native, data-driven, and distributed systems, with a track record in digital transformation across energy, public sector and applied AI. Experienced in leading work from discovery through to production, balancing strategic intent with pragmatic engineering.

Experience

Data Engineer - Simpson Associates (Short-Term Contract)

Oct 2025 - Dec 2025 Short-term engagement delivering PySpark ingestion pipelines and Azure Fabric medallion architectures. Produced Power BI reports and star-schema dimensional models used across multiple client projects.

Consultant Software Engineer - San Digital

July 2022 - October 2025

Summary Founding engineer at a startup consultancy, one of a small senior team trusted to lead client engagements end-to-end. Played a hands-on role in shaping engineering culture, technical direction, and delivery practices as the company scaled. Regularly operated as de facto technical lead, balancing hands-on delivery with architecture, stakeholder engagement, and mentoring.

Key Achievements

- One of the first engineers hired, helping establish the consultancy's engineering culture, delivery standards, and core technical frameworks adopted across client engagements.

- Provided technical leadership across projects in energy, reservations, mental health, and applied AI, making architectural decisions appropriate to client maturity, scale, and risk.
- Delivered production systems from MVP through to live operation, combining hands-on coding with architectural oversight across web, mobile, and cloud platforms.
- Acted as primary technical interface for clients, refining requirements, shaping roadmaps, running demos, and aligning technical decisions with business strategy.
- Led technical input for R&D tax credit claims, documenting and articulating innovative engineering work in areas such as AI, automation, and distributed systems.
- Represented the consultancy externally, co-creating and presenting LinkedIn video content explaining technical concepts to non-technical audiences.
- Co-authored a well-received AI book for engineers and senior leaders, providing practical guidance on application, risk, and strategy.

Selected Projects

- EV Charger Reservations - Designed and delivered an event-driven platform managing reservations for limited EV charging capacity. Integrated real-time vehicle arrival detection via number plate recognition to automatically activate reservations. Built reliable workflows for allocation, conflict handling, retries, and monitoring across distributed Azure services.
- Energy sector - Designed and delivered customer-facing platforms for purchasing solar panels and EV chargers, including complex financing and loan-based purchasing flows.
- Mental health - Designed and built a mobile application and backend for continuous patient well-being tracking, replacing manual questionnaires with higher-frequency assessment.
- AI and Algorithm initiatives - Developed applied AI tooling including generated-text detection, intelligent icon layout, and patent summarisation and categorisation against technical standards.

Tech/Skills TypeScript, React, Node.js, Python, AWS, Azure, solution architecture, agile delivery, requirements gathering, demos.

Principal Engineer/Tech Lead - BJSS

October 2017 - July 2022

Summary Consultant and technical lead at a major agile-focused software consultancy, delivering complex programmes for high-profile clients across government, finance, and legal sectors. Balanced hands-on engineering with architec-

ture, leadership, and senior client-facing responsibilities. Involved across the full delivery lifecycle including bids, pre-sales, discovery, delivery, and live support.

Key Achievements

- Acted as technical lead on award-winning government digital services, transforming traditional paper based services to digital in line with GDS standards. Recognised internal expert on GDS alpha-beta-live delivery, supporting teams adopting the standards for the first time.
- Shaped client cloud adoption strategies across AWS and Azure, helping organisations modernise infrastructure while balancing cost, performance, and operational risk.
- Led architecture and development for high-risk systems, including:
  - A currency exchange platform with finite availability under heavy traffic.
  - A patent allocation platform requiring fair, auditable round-robin assignment.
- Led and mentored development teams, providing technical leadership, coaching, and formal line management.
- Security advocate for the Nottingham office: ran OWASP-based interactive security training, supported secure code reviews across projects, and educated teams on JWT and common vulnerability patterns.
- Strong agile practitioner, working closely with BAs, delivery managers, and clients to shape backlogs and communicate progress clearly to both technical and non-technical stakeholders.
- Played a key role in scaling BJSS regional offices (Nottingham, Birmingham, Lisbon), contributing to competency-based interviews and graduate academy programmes.
- Represented BJSS externally, winning the Hack24 Microsoft Azure category for innovative use of cognitive services.

Selected Projects

- Government services - Technical lead building web portals and mobile apps for vehicle accounts and booking management; technical lead for the alpha stage of an award-winning digital transformation in the transport sector (Ionic/TypeScript/AWS).
- Chatbots - Designed and implemented an internal serverless chatbot framework, later deployed to a client to reduce contact centre load (AWS/JavaScript/Python).
- Currency exchange - Tech lead migrating a legacy monolith into distributed microservices (C#/Service Fabric).
- Finance (Multiple) - Data engineering platforms for large-scale processing and analytics (Python/PySpark/Azure; Python/Pandas/AWS).
- Law - Tech lead with a law firm on a patent renewal architecture and delivered the resulting client portal and scheduling services

(C#/Angular/Azure).
- Consulting engagements - Advised a manufacturing firm on system landscape and defined a practical modernisation roadmap.

Tech/Skills TypeScript, Ionic, Angular, React, Node.js, Python, C#, Service Fabric, AWS, Azure, microservices, serverless architecture, data engineering, agile delivery, stakeholder workshops, line management, interviewing & hiring.

Software Developer/Team Lead - Romax Technology/Romax InSight

January 2014 - September 2017

Summary Promoted to team lead in a growing software division that later became ONYX Insight, a Castrol/BP company. Led a team of three developers to deliver a global wind turbine monitoring platform, while modernising the codebase, architecture, and delivery practices. Balanced hands-on coding, architectural design, and mentoring with responsibility for delivery during a major acquisition.

Key Achievements

- Delivered a global wind turbine condition monitoring system, enabling asset owners worldwide to track performance and prevent failures.
- Successfully managed an audit and the technical handover during acquisition by Castrol, ensuring continuity of systems, knowledge transfer, and team alignment.
- Transitioned a legacy monolithic frontend into a modern Angular application, introducing a microservices-based architecture.
- Re-engineered the delivery pipeline, moving from 6-9 month releases to a Kanban-based approach with weekly releases when required.
- Collaborated closely with mechanical and electrical engineers, converting MATLAB models and data analysis into production-ready visualisations within the monitoring platform.
- Reverse-engineered proprietary data formats to ensure interoperability with customer systems and reliable data ingestion.
- Co-developed an in-house vibration monitoring hardware kit, writing embedded C code and integrating it into the central monitoring platform. Supported the electrical engineering team with Git workflows and best practices.
- Worked with a distributed team (UK, US, Korea), coordinating requirements and delivery across time zones.

Selected Projects

- Wind turbine monitoring platform - Python backend with JavaScript frontend; integrated customer data streams into centralised systems used

globally.
- Engineering integration - Translated MATLAB-based analysis into graphs and dashboards for end users; reverse-engineered turbine data files to support new clients.
- Custom vibration monitoring kit - Contributed C code for embedded hardware and integrated the resulting data streams into the main monitoring system.
- System modernisation - Migrated legacy codebase to microservices with REST APIs; upgraded frontend to Angular.

Tech/Skills Python, JavaScript, Angular, REST APIs, AWS (EC2, S3), PostgreSQL, Linux, microservices, agile/Kanban, embedded C, data visualisation, reverse engineering, acquisition handover, technical mentoring.

Senior Software Developer - Nikon Metrology UK

February 2010 - December 2013

Worked on Camio, Nikon's flagship CAD-oriented metrology software, used in aerospace and automotive industries by organisations such as Volkswagen, Jaguar, and Boeing. Developed complex C++ systems controlling industrial robotic arms, lasers, and touch probes for micron-level precision measurement. Took ownership of major subsystems and contributed to modernising a 20-year-old codebase.

Key Achievements

- Redesigned and upgraded the data model and database layer, migrating from MSDE 2000 to SQL Server and introducing a templated ADO wrapper with caching.
- Improved software reliability and maintainability by introducing Boost libraries and unit testing for critical features.
- Delivered a major feature upgrade for multi-column measurement, coordinating multi-process synchronisation across robotic arms.
- Built standalone database administration tools in C# to support on-site engineers and streamline client deployments.
- Supported industrial engineers with installation documentation and troubleshooting for secure, locked-down systems.
- Future-proofed the product by resolving compatibility issues with new Windows versions and preparing the codebase for 64-bit.
- Worked with a distributed team (UK, US, Belgium), including travel for collaboration and knowledge sharing.

Selected Projects

- Database modernisation - Migrated Camio's database backend, delivering performance improvements and maintainability gains.

- Robotics synchronisation - Designed and implemented multi-column measurement capability requiring advanced process coordination.
- Tooling for engineers - Built supporting utilities in C# to simplify database administration and customer support workflows.

Tech/Skills C++, STL, MFC, Boost, COM, C#, SQL Server (2008 R2), T-SQL, Subversion, Jira, robotics.

Software Developer - DriveSentry Inc.

August 2005 - November 2009

Founding engineer at a data protection startup, developing consumer-facing antivirus and encryption software in C++/C#. Contributed to UI, rules engine, and backend systems that scaled to tens of thousands of daily transactions. Product reached retail shelves (PC World) and magazine cover disks (PC Advisor).

Tech/Skills: C++, C#, SQL Server, Windows internals, XML, SOAP

Junior Software Developer - Bitarts

December 2004 - July 2005

Built a PHP/Oracle web frontend for online software activation at a security software firm specialising in DRM and anti-piracy solutions.

Tech/Skills: PHP, Oracle, C++, Solaris

Founder / Independent Developer - Crumbly Head Games

2010 - Present (part-time) Built, maintain and sell a desktop product to help authors design and visualise interactive fiction and choose your own adventure games. Responsible for end-to-end everything: design, coding, marketing, sales, and customer support. Currently developing a new online/Rust version.

Tech/Skills: C++, MFC, Boost, GraphViz, Crypto++, Rust

Earlier Programming Work

Developed small websites and independent C++/DirectX games prior to entering full-time employment.

Education

- BSc Computer Science (2:2) - Nottingham Trent University, 2001-2003
  Final year dissertation: Simulating Genetic Inheritance
- HND Computer Studies (Merit) - West Nottinghamshire College, 1999-2001
- A-Levels & GCSEs - The Manor Comprehensive School, 1992-1999
  Maths, Biology, Physics, General Studies